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**Bennett**

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(54) **SIDE UNDERRIDE GUARD**

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(22) Filed: **Feb. 21, 2025**

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(51) **Int. Cl.**
***B60R 19/56*** (2006.01)
***B62D 65/16*** (2006.01)

(52) **U.S. Cl.**
CPC ............ ***B60R 19/565*** (2013.01); ***B62D 65/16*** (2013.01)

(58) **Field of Classification Search**
USPC ........................................ 296/180.4; 293/128
See application file for complete search history.

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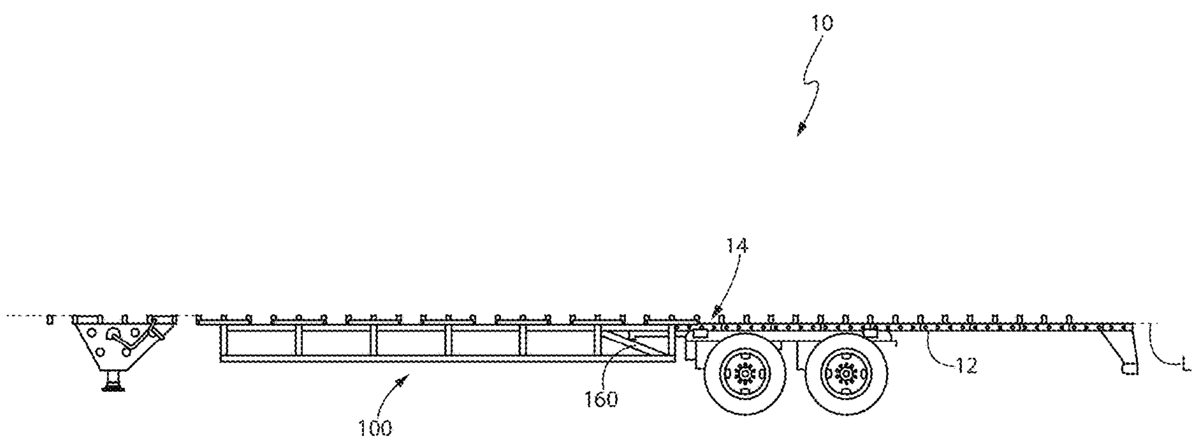
(57) **ABSTRACT**

A side underride guard for vehicles, including trailers for tractor-trailer rigs, having a series of triangular shaped stabilizers made from a vertical section extending downwardly from the structural siderail of the trailer, a horizontal section extending from the top of the vertical section, and a diagonal section extending from the bottom of the vertical section to an inboard side of the horizontal section. A series of these stabilizers are mounted along two horizontal beams to form a triangular prism, but do not attach to crossmembers of the trailer. A pair of side underride guards are positioned on opposite sides of the trailer bed in which the two side underride guards act independently of each other.

**25 Claims, 11 Drawing Sheets**



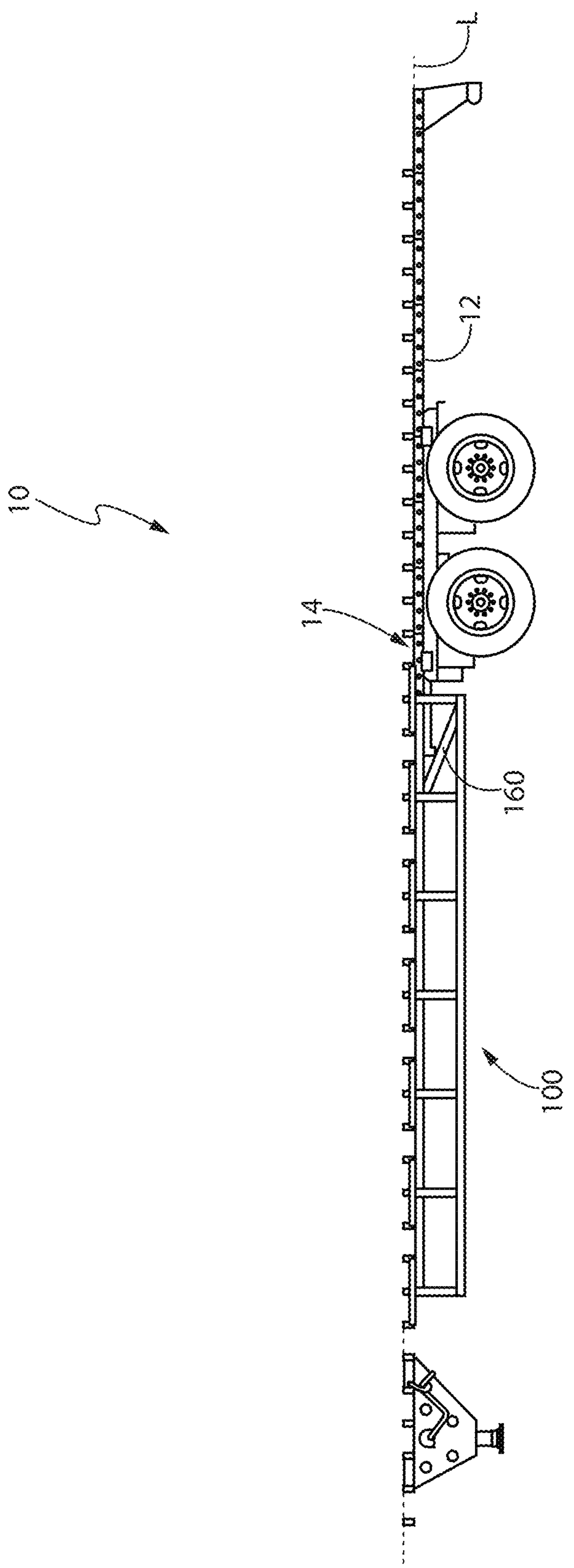



FIG. 1

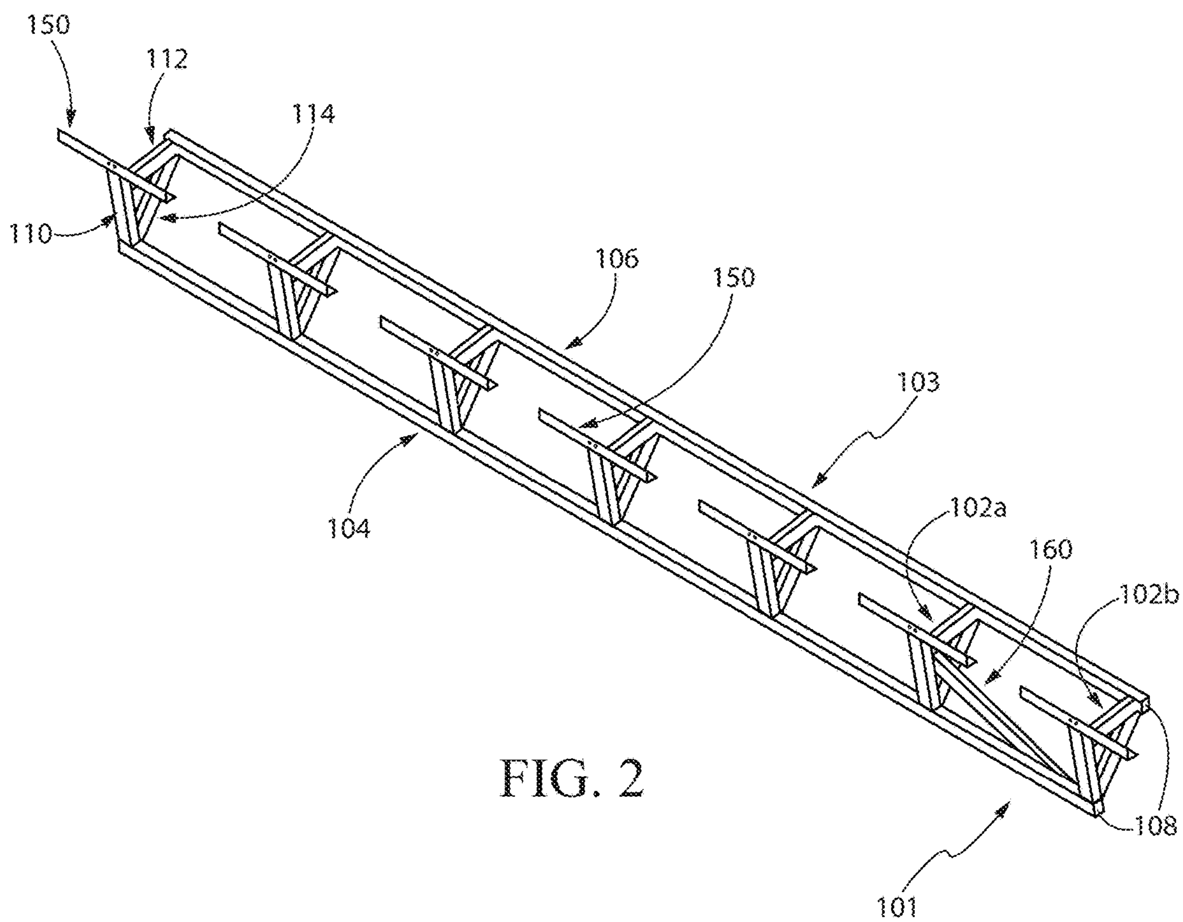
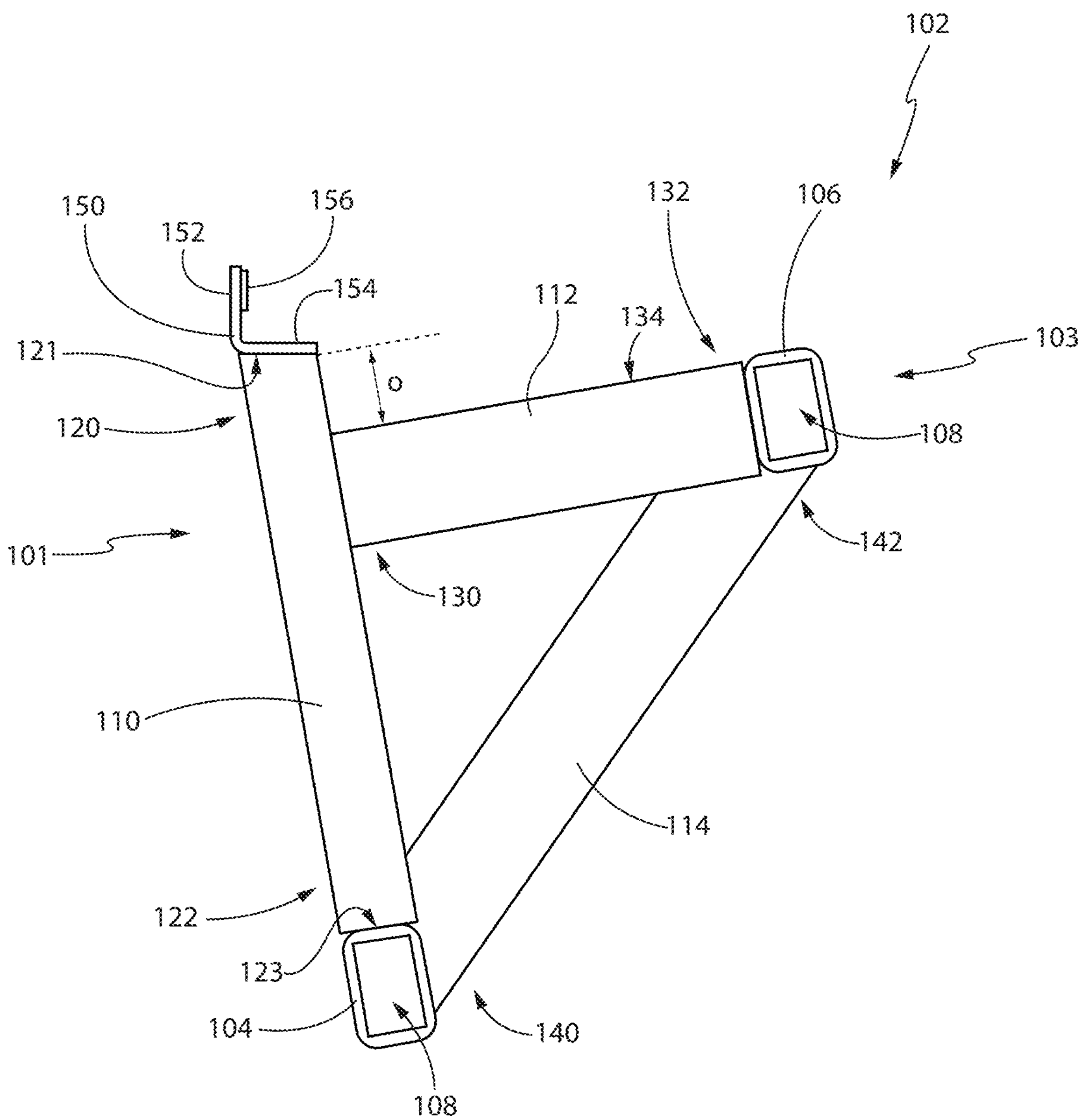



FIG. 2


102
132
106
150
156
152
154
112
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103
121
o
120
108
142
101
130
110
114
122
123
104
140
108

FIG. 3

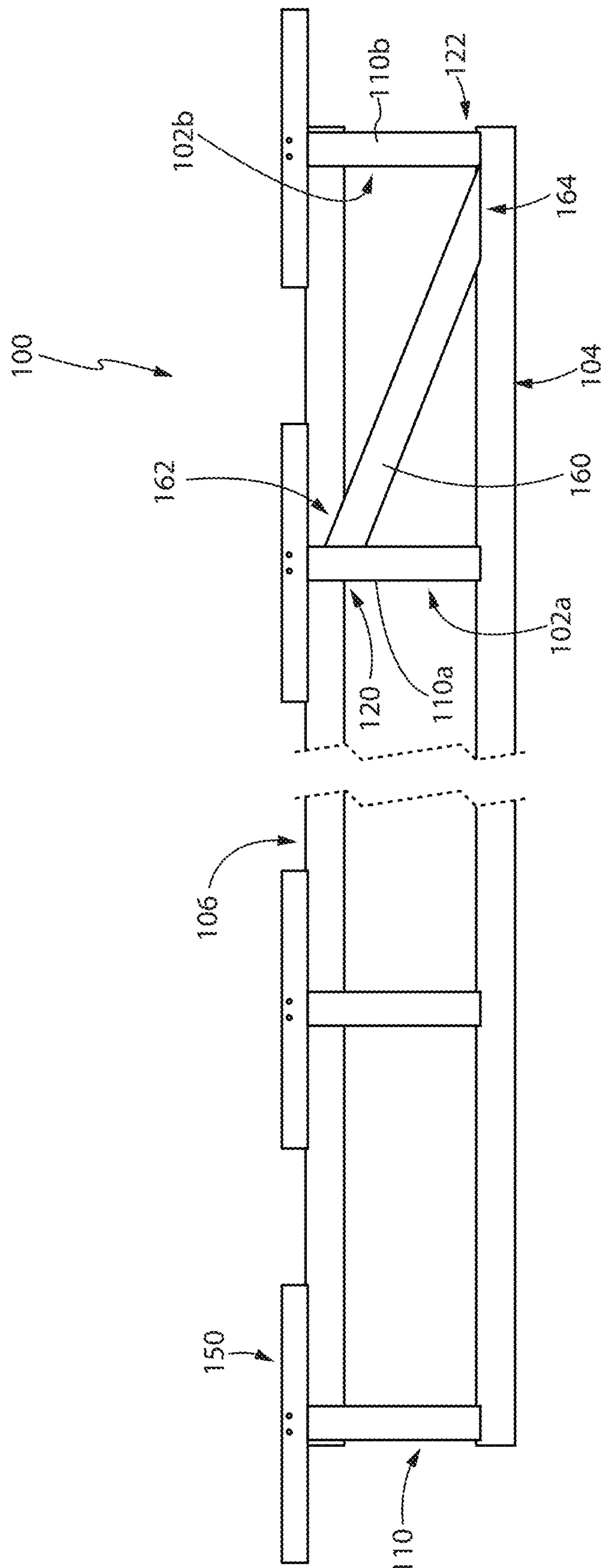



FIG. 4

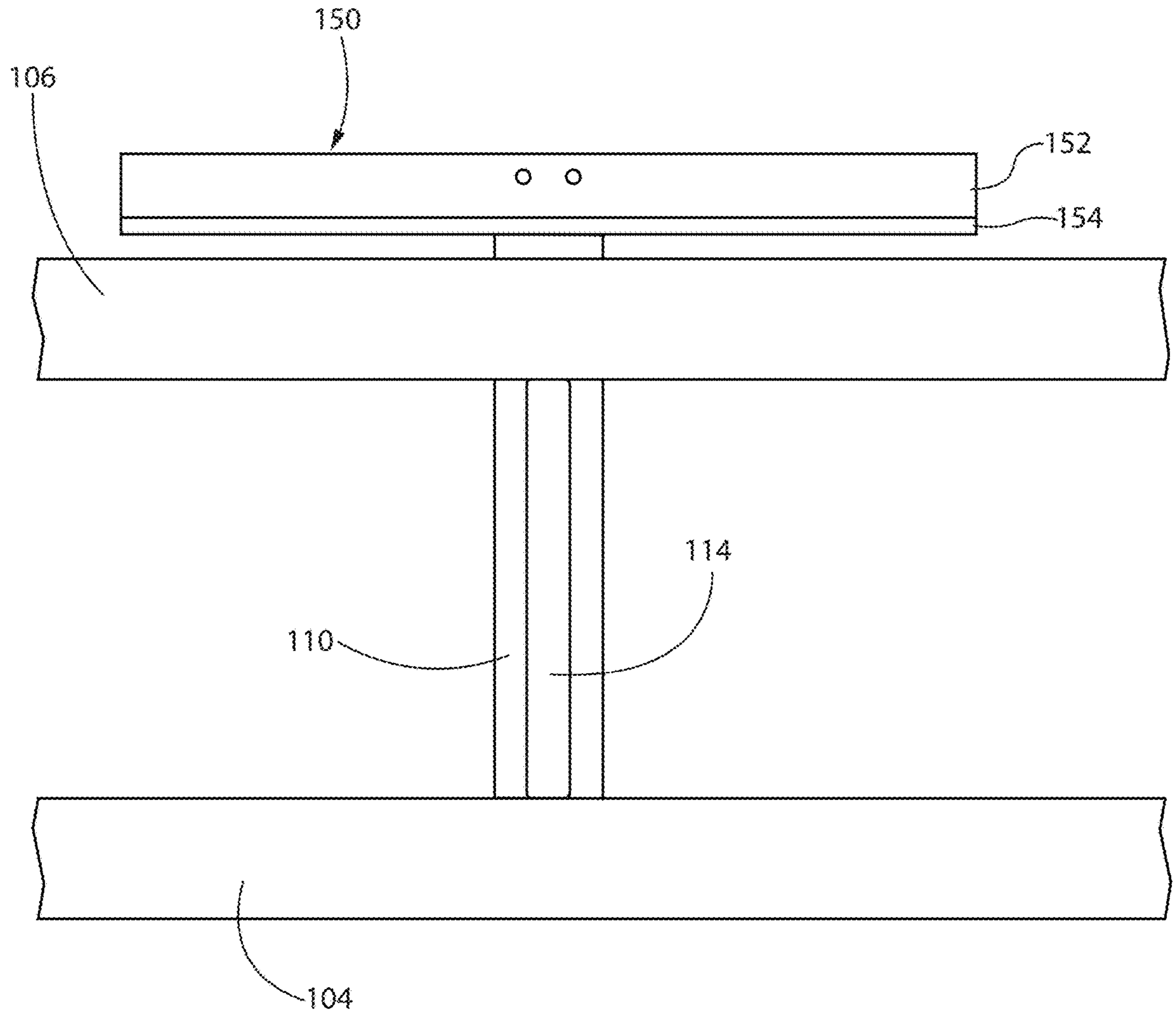
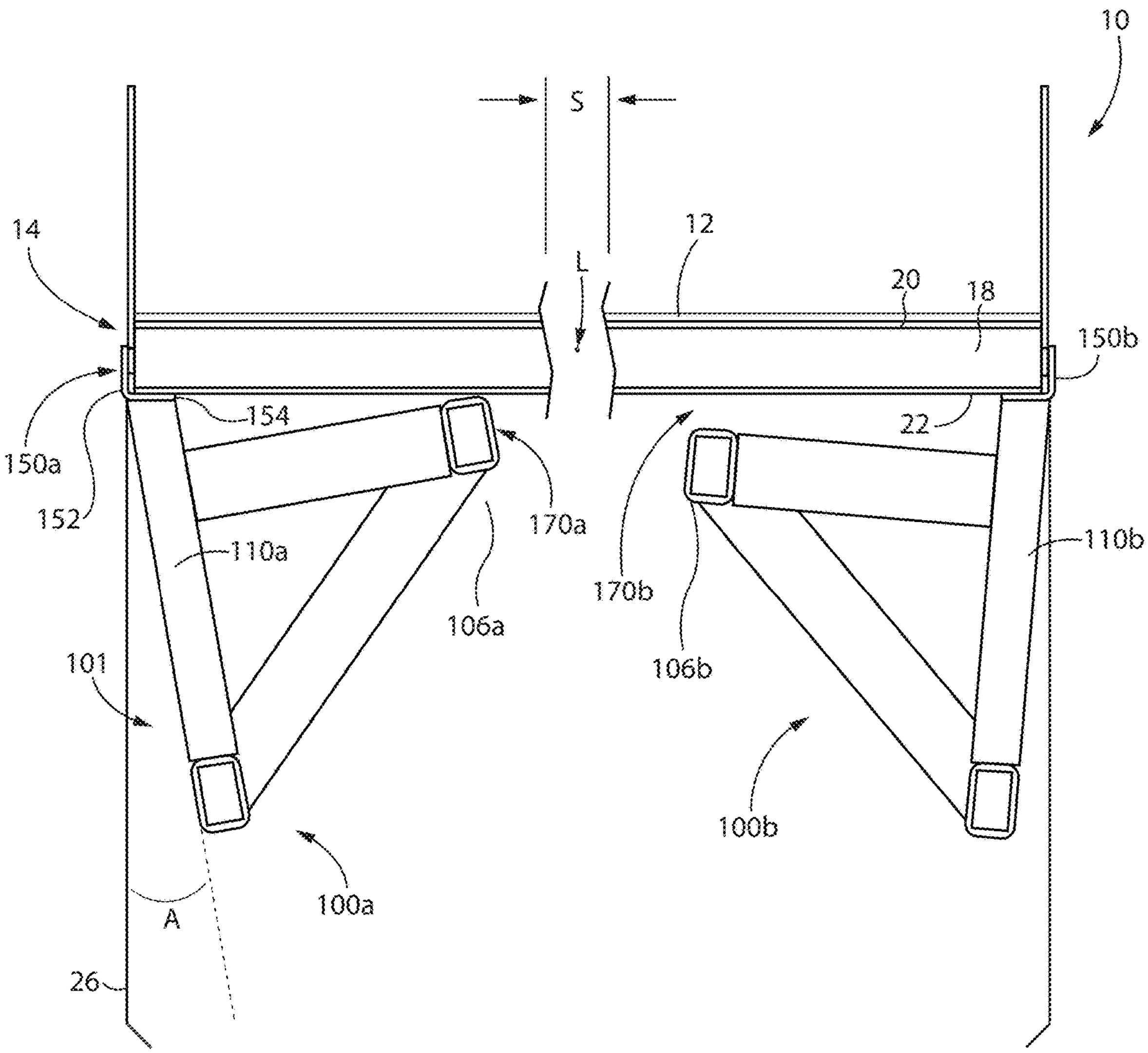



FIG. 5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S
14
12
L
20
18
150b
150a
154
22
152
170a
110a
170b
110b
106a
106b
101
100b
A
100a
26

FIG. 6

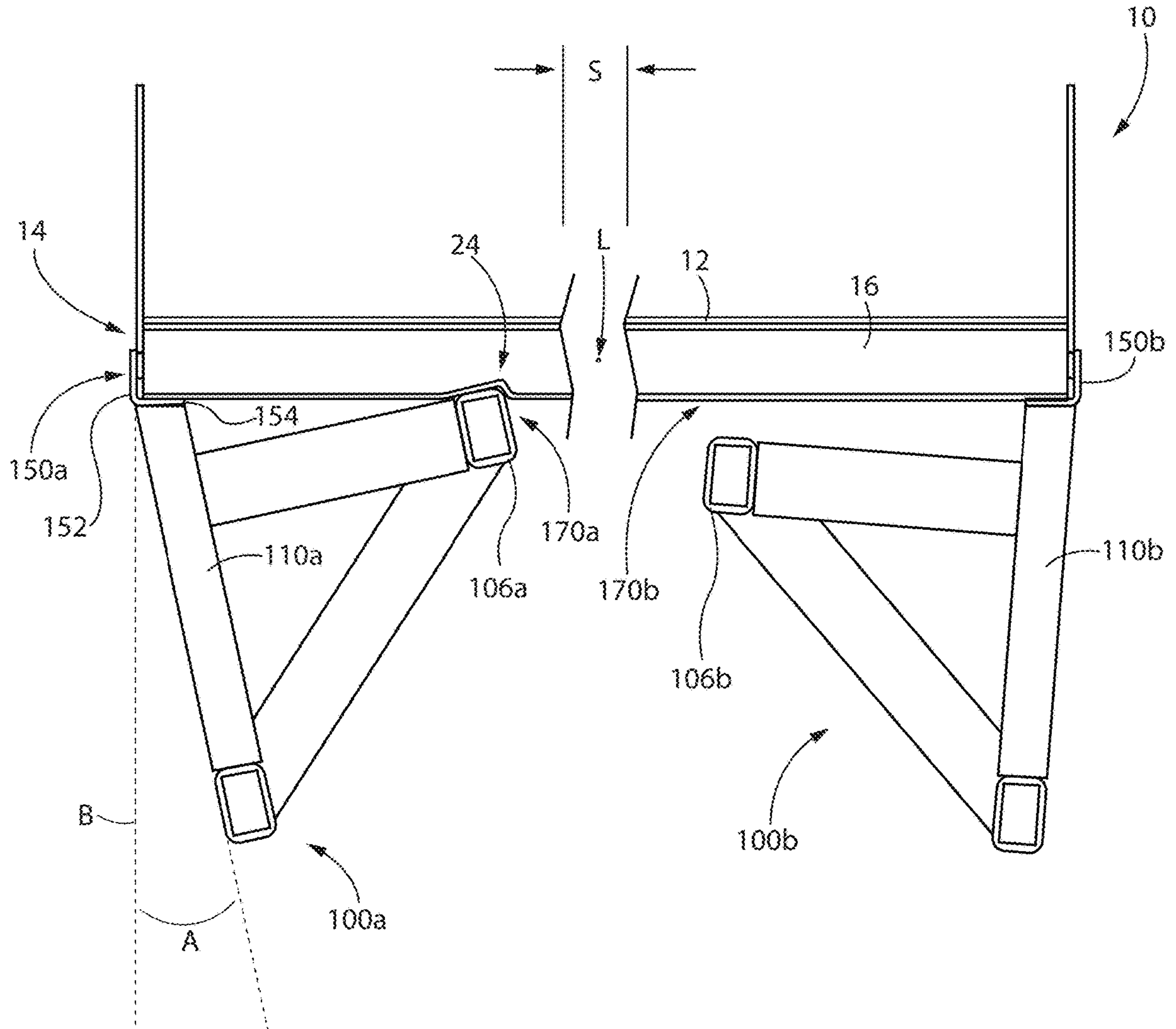
10
S
14
24
L
12
16
150b
154
150a
152
110a
170a
106a
170b
110b
106b
B
100b
A
100a

FIG. 7

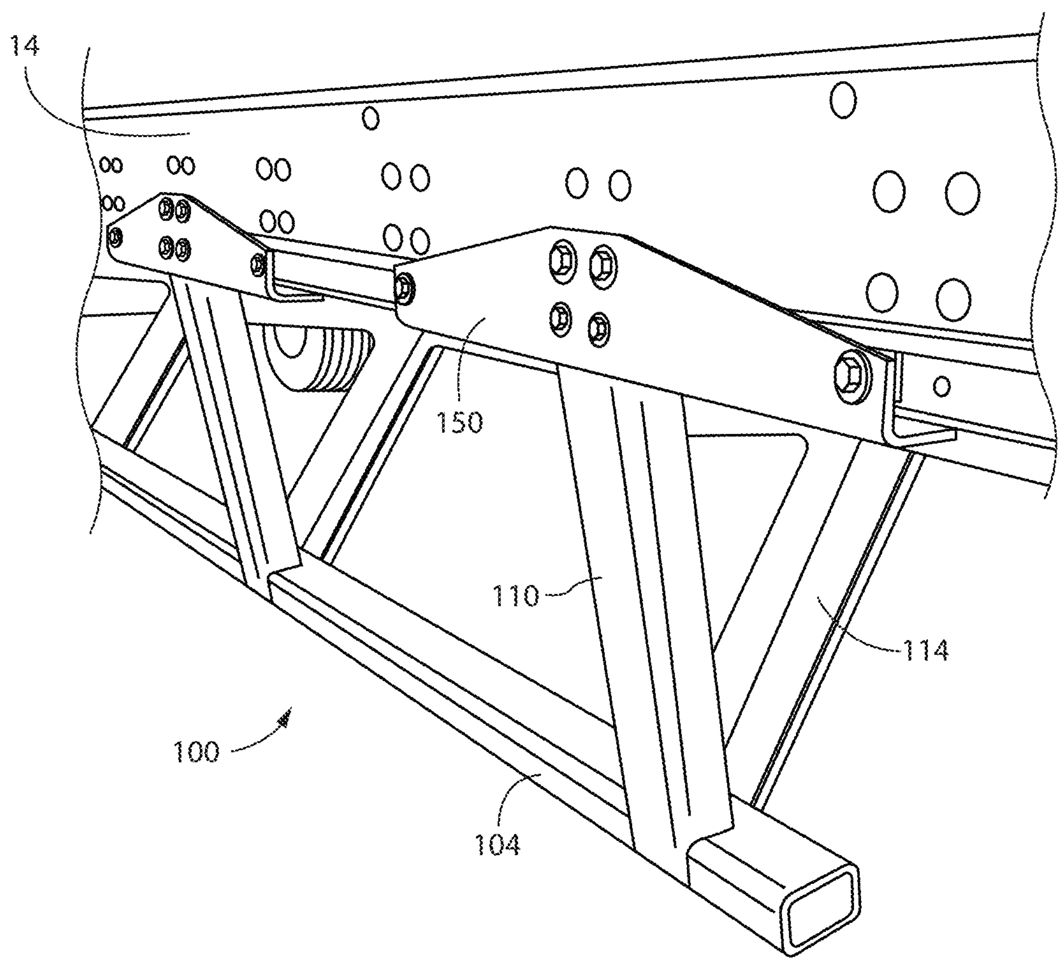


FIG. 8

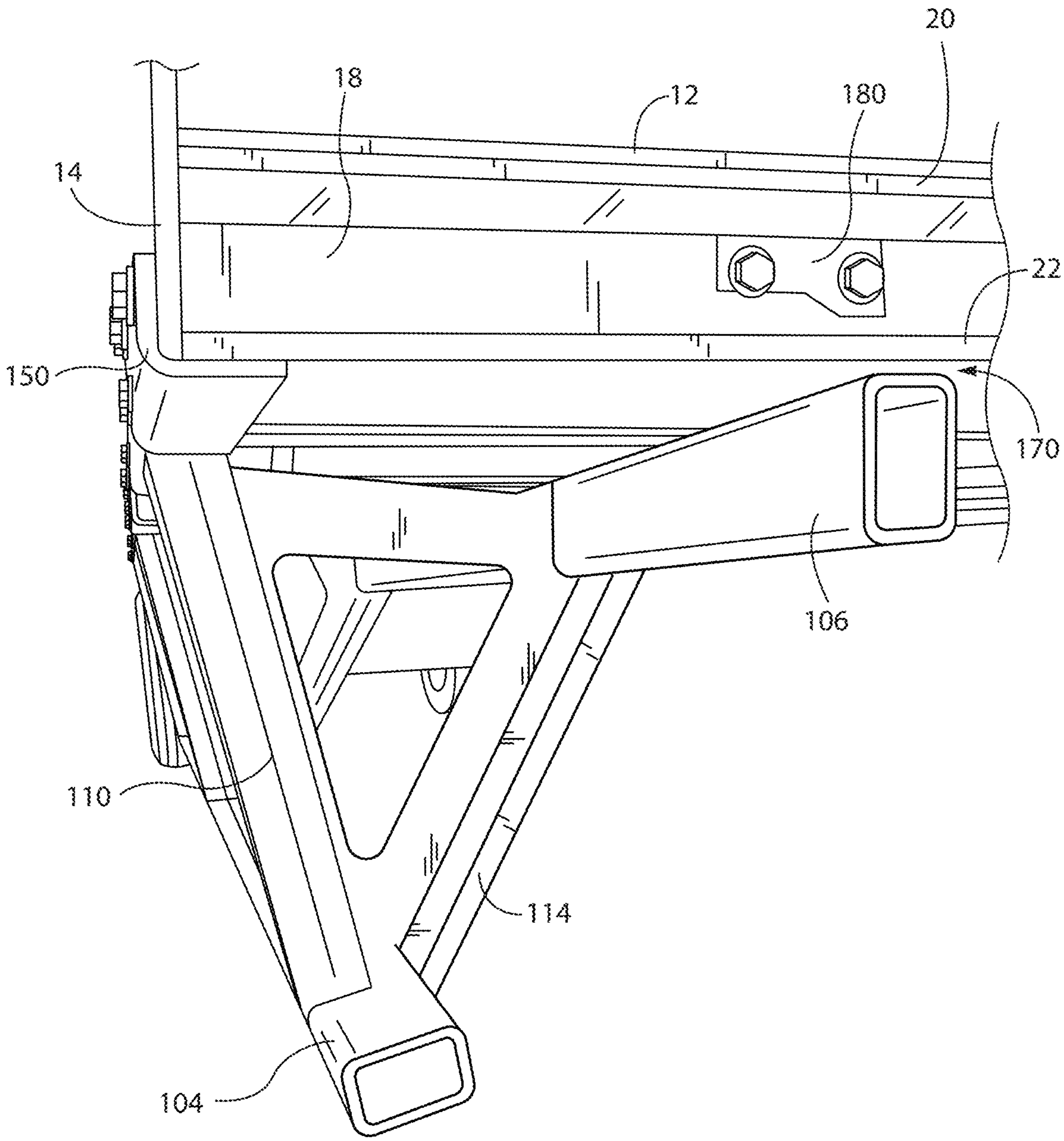


FIG. 9

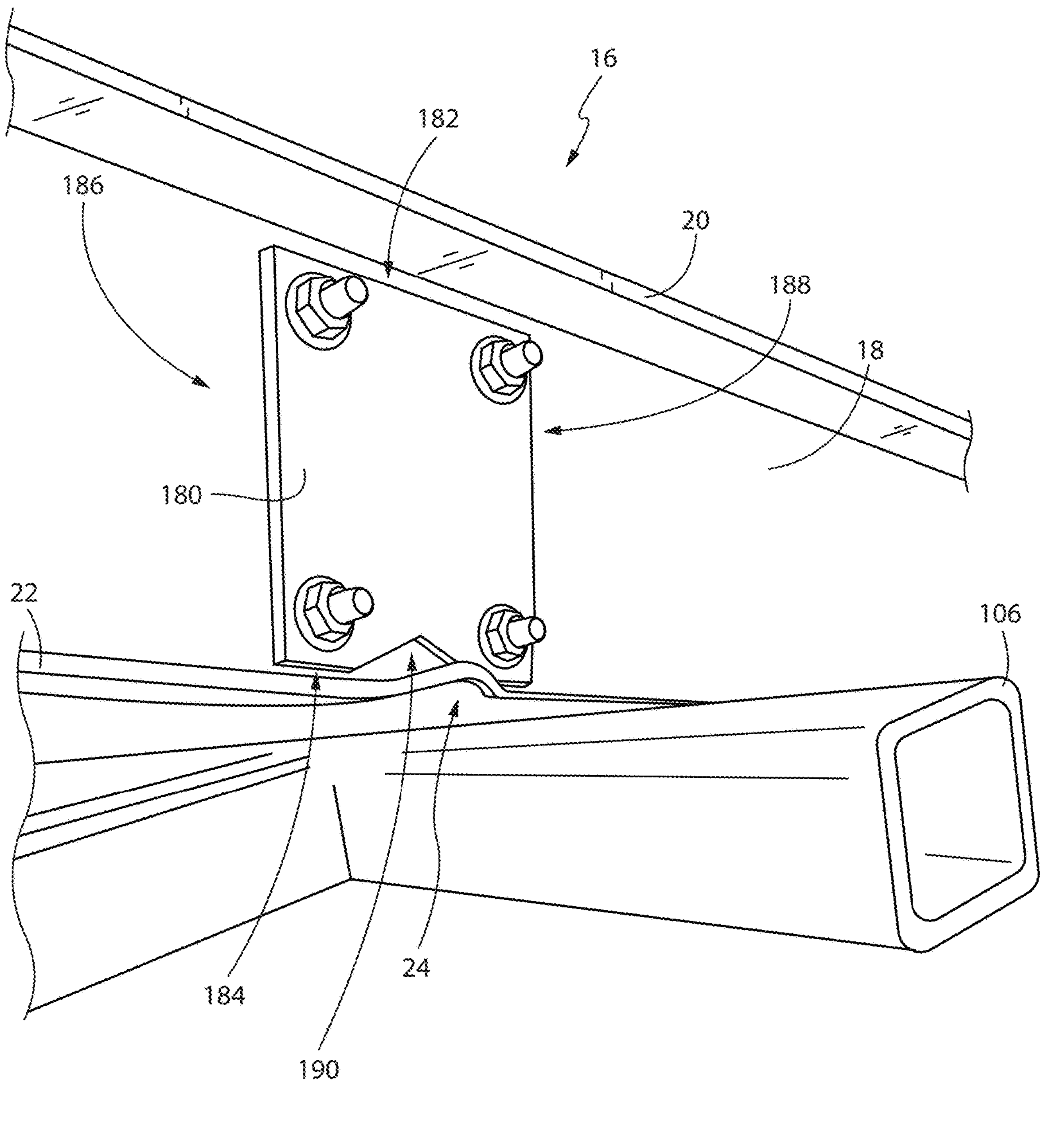


FIG. 10

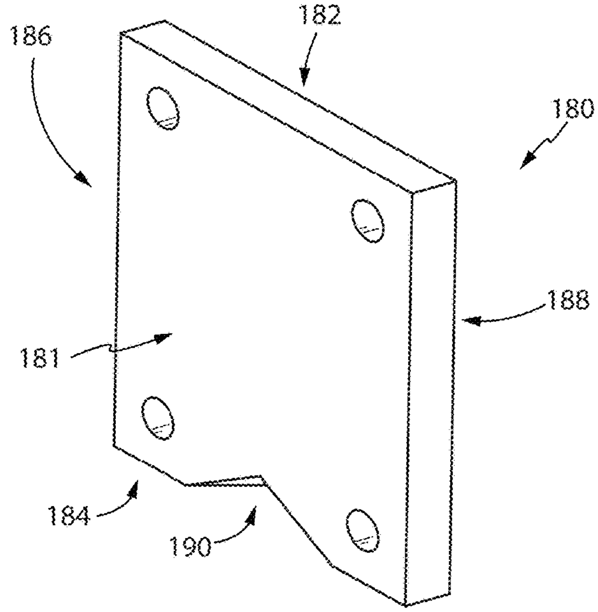


FIG.11A

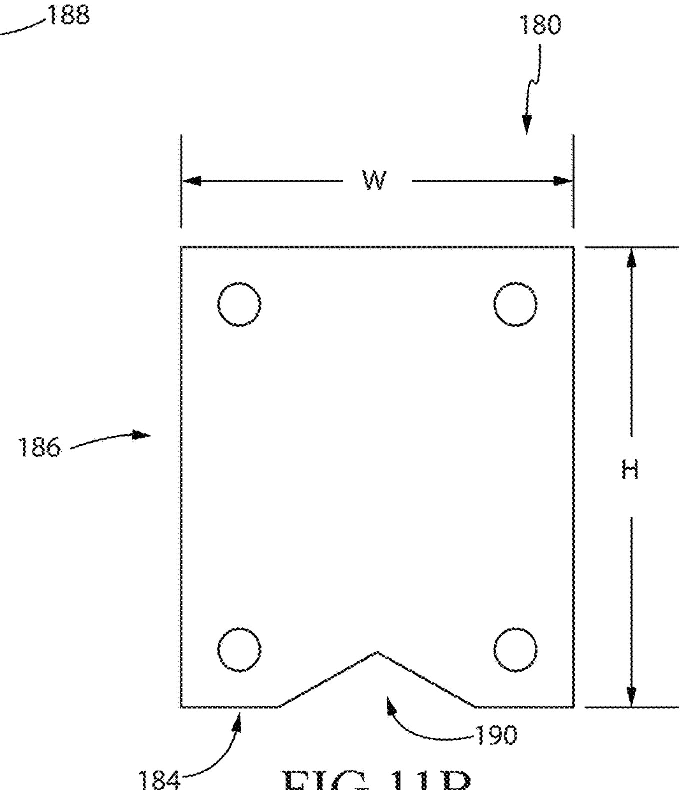


FIG.11B

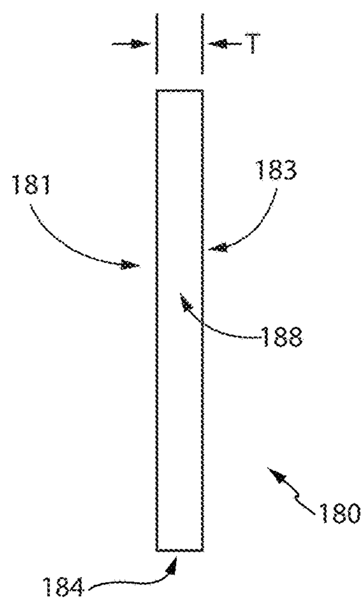


FIG.11C

# SIDE UNDERRIDE GUARD

## CROSS REFERENCE TO RELATED APPLICATION

This patent application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/556,257, entitled "Side Underride Guard," filed Feb. 21, 2024, which application is incorporated in its entirety here by this reference.

## TECHNICAL FIELD

This invention relates to vehicle accident mitigation.

## BACKGROUND

Rear Underrun Protection Systems known as underride guards are rigid assemblies at the rear of vehicles including trailers and semi-trailers. The Federal Department of Transportation has mandated these assemblies through NHTSA REGULATIONS FMVSS **223** and **224** with the intention that the bars will interfere with automobiles running under such high, heavy vehicles in a rear end collision. The object of the Federal Regulations is to reduce the likelihood of impact of the vehicle frame with the automobile passenger compartment. The rigid assemblies typically include vertical brackets depending from the vehicle to support a horizontal bar extending the width thereof. The length of the bars and ground clearance are dictated by the regulations to receive the structural areas of an automobile bumper and crumple zone. The mandated clearance is enabled by the proximity of the guards to the rear wheels, making ground contact of the horizontal bar highly unlikely.

Side underride guards have not been mandated by NHTSA regulations even though there is some possibility of side underride from automobiles intersecting such high, heavy vehicles from substantial angles. Of course, nothing can avert predictable disaster at extreme impact speeds. With side underride guards, the guards, by definition, would extend to a significant distance between the vehicle front and rear wheels. Such side underride guards adhering to the NHTSA regulations for rear underride guards would make any trailer subject to recurrent high centering during normal operation. The problem of high centering is regularly encountered with aerodynamic side skirts on such vehicles, requiring a compromise between side skirt flexibility and aerodynamic effectiveness. U.S. Pat. No. 11,691,583, incorporated here in its entirety, solves this problem. However, this underride guard adds significant weight and structure to the trailer.

For the foregoing, there is a need for improved underride guards that reduce the added weight to the trailer without compromising structural integrity and safety.

## SUMMARY

The present invention is directed to side impact guard systems, such as a side underride guard for vehicles, including trailers for tractor-trailer rigs. In a first aspect of the present invention, the side underride guard has an effective ground clearance on flat ground that is above bumpers and tires of the standard automobile as defined by the NHTSA but within the tire lift zone of wheel wells and below the top of structural elements at the back of the engine compartment. As a result, the body of the automobile and its frame



below the windshield will reduce the likelihood of intrusion of the vehicle frame into the automobile passenger compartment.

The side underride guard includes vertical sections extending downwardly from the structural siderail of the trailer. A horizontal beam extending longitudinally of the vehicle is fixed below the vehicle structural siderail to the lower ends of the vertical sections. The horizontal beam is to have a nominal effective ground clearance of about twenty-seven inches.

In a second aspect of the present invention, a pair of side underride guards are positioned on opposite sides of the floor of the trailer in which the two side underride guards act independently of each other.

In a third aspect of the present invention, each side underride guard includes a series of stabilizers mounted along a horizontal beam to form a triangular prism shape.

In a fourth aspect of the present invention, the side underride guard includes lateral bracing extending between the structural siderails and the horizontal beam. The lateral bracing includes longitudinal braces fixed at one end to one of the structural siderails and to the horizontal beam at the lower end of the vertical sections at the other end, respectively. By having the bracing extend to the structural siderails, stress raisers in the trailer bed are mitigated upon an impact on the guard bar, which can occur if support is affixed to the trailer bed. Such stress raisers would compromise the flexibility of the vehicle bed to adjust under cargo loads.

Accordingly, it is an object of the present invention to provide an improved side underride guard. Other and further objects and advantages will appear hereafter.

## BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** shows a side elevation view of a trailer with an embodiment of the side underride guard installed.

FIG. **2** shows a perspective view of an embodiment of the side underride guard.

FIG. **3** shows a rear view of an embodiment of the stabilizer.

FIG. **4** shows a side elevation view from the outboard side of an embodiment of the side underride guard.

FIG. **5** shows a side elevation view from the inboard side of an embodiment of a portion of the side underride guard along a stabilizer.

FIG. **6** shows a rear view of the present invention in its natural state before any lateral impact.

FIG. **7** shows a rear view of the present invention showing the effects of lateral impact.

FIG. **8** shows a perspective view of a portion of the side underride guard installed on a trailer.

FIG. **9** shows a perspective view of a portion of the side underride guard installed on a trailer as viewed longitudinally down the centerline of the trailer.

FIG. **10** shows a partial view of the side underride guard after lateral impact causes the side underride guard to form a saddle in the floor-support crossmember.

FIGS. **11**A-**11**C show perspective, front, and side views of the reinforcing plate, respectively.

## DETAILED DESCRIPTION OF THE INVENTION

The detailed description set forth below in connection with the appended drawings is intended as a description of presently-preferred embodiments of the invention and is not intended to represent the only forms in which the present invention may be constructed or utilized. The description sets forth the functions and the sequence of steps for constructing and operating the invention in connection with the illustrated embodiments. It is to be understood, however, that the same or equivalent functions and sequences may be accomplished by different embodiments that are also intended to be encompassed within the spirit and scope of the invention. In addition, directional terminology is not intended to be limiting, but rather used for ease, efficiency, and clarity. As such, directional terms such as top, bottom, upper, lower, vertical, horizontal, diagonal, and the like, are used in reference to the present invention being properly installed on a trailer and viewing the trailer along its longitudinal centerline.

With references to the figures, a trailer **10** for tractor-trailer rigs is illustrated in FIG. **1**. This trailer **10** may be a vehicle with trucks at each end or only at the rear end in the form of a semi-trailer, either being associated with a tractor-trailer rig. Both such vehicle configurations are here referred to as a trailer. Such trailers are high and heavy and automobiles can underride the trailer in a collision. The trailers **10** generally comprise a trailer bed that defines the floor **12** of the trailer **10**. The floor **12** is generally flat and rectangular in shape defining a longitudinal centerline L approximately central to the opposing long sides of the floor **12** and along the length of the trailer. In general, the floor **12** of the trailer **10** is supported by floor-support crossmembers **16** that span the width of the floor **12**. A plurality of floor-support crossmembers **16** can be positioned intermittently spaced apart along the length of the floor **12**. The floor-support crossmembers **16** can be in the form of an I-beam having a web **18** terminating at opposite flanges, referred to herein as a top flange **20** and a bottom flange **22** based on proper installation. Properly installed, the top flange **20** is in direct contact with the floor **12** as shown in FIG. **6**.

The trailer **10** has a side impact guard, in the form of a side underride guard **100** that runs parallel to the longitudinal centerline L of the trailer **10** to prevent automobiles from riding too deep under the floor **12** of the trailer **12** during a side collision. The side underride guards **100** are attached to structural siderails **14** on the outboard side of the the floor **12**. As such, the side underride guards **100** have an outboard side **101** configured to attach to the siderails **14** at the sides of the floor **12**, and an inboard side **103** directed towards the longitudinal centerline L of the floor **12**. In the preferred embodiment, the side underride guards **100** do not attach to the floor-support crossmembers **16** of the trailer **10**. In addition, when installed, the side underride guard **100** should not pass the longitudinal centerline L. The side underride guard **100** reduces the likelihood of an automobile driven into the side of the trailer **10** continuing thereunder such that the passenger area of the automobile reaches the structural siderails **14** of the frame of the trailer **10**.

With reference to FIG. **2**, the side underride guard **100** comprises a series of repeating stabilizers **102** arranged in a series along two beams **104**, **106**. The beams **104**, **106** can be constructed as a hollow tube, preferably, made of metal. End caps **108** can be placed at the ends of the beams **104**, **106** to cover the tube. Each stabilizer **102** is generally triangular in shape as viewed from the rear, not the side, of the trailer (i.e. along the centerline L). As such, the underride guard **100** generally has an overall triangular prism shape. The triangular shape of each stabilizer **102** is defined by three sections. Based on the proper installation of the side underride guard **100** on a trailer **10** and viewed from the back to the front, the first leg is referred to as a vertical section **110**, the second leg is referred to as a horizontal section **112**, and the third leg is referred to as a diagonal section **114**. In the preferred embodiment, the vertical section **110** and the horizontal section **112** form approximately a right angle, making the diagonal section **114** the hypotenuse. The vertical section **110**, horizontal section **112**, and diagonal sections **114** can be formed from metal plates, tubes, formed angles, channels, and the like. The form of these sections can depend on the material used. For example, if lighter weight aluminum is used, then the vertical section **110**, horizontal section **112**, and diagonal section **114** can be made of tubes. If heavier weight steel is used, then the vertical section **110**, horizontal section **112**, and diagonal section **114** can be formed as angles or channels to reduce weight without compromising strength. In some embodiments, a combination of tubes, channels, angles, and the like can be used to form the sections.

In some embodiments, the vertical section **110**, the horizontal section **112**, and the diagonal section **114** can be separate pieces that are connected together (for example, via welding or any other fastening technique) to form the triangle. In some embodiments, the vertical section **110**, the horizontal section **112**, and the diagonal section **114** can be integrally formed from a single piece of material. For example, the vertical section **110**, the horizontal section **112**, and the diagonal section **114** can be vertical, horizontal, and diagonal flanges of one single part, and formed by sheet metal pressing or casting from the same single part or flat sheet of metal.

With reference to FIG. **3**, the first leg or vertical section **110** comprises an upper end **120** terminating at a top surface **121**, and a lower end **122** terminating a bottom surface **123** opposite the upper end **120**. The second leg or horizontal section **112** comprises an outboard end **130** and an inboard end **132** opposite the outboard end **130**. The third leg or diagonal section **114** comprises a lower-outboard end **140** and an upper-inboard end **142** opposite the lower-outboard end **140**. In some embodiments, the outboard end **130** of the horizontal section **112** can be operatively connected to the upper end **120** of the vertical section **110**, the lower-outboard end **140** of the diagonal section **114** can be operatively connected to the lower end **122** of the vertical section **110**, and the inboard end **132** of the horizontal section **112** can be operatively connected to the upper-inboard end **142** of the diagonal section **114**, thereby forming the triangular shape.

In the preferred embodiment, the outboard end **130** of the horizontal section **112** is offset slightly from the top surface **121** of the upper end **120** of the vertical section **110**. For example, the top side **134** of the horizontal section **112** can have an offset distance O of about at least about 1 inch to about 5 inches (and preferably about 2.125 inches) below the top surface **121** of the upper end **120** of the vertical section **110**.

With reference to FIGS. **2-5**, the lower end **122** of the vertical section **110** and/or the lower-outboard end **140** of the diagonal section **114** can be operatively connected to the first beam **104** in approximately a perpendicular manner to the first beam **104** when viewed from the outboard side **101** or the inboard side **103** as shown in FIG. **5**. When the underride guard **100** is properly installed on a trailer **10**, the first beam **104** is positioned on the outboard side **101** of the underride guard **100** in a horizontal orientation extending parallel to the longitudinal centerline L of the floor **12**. The first beam **104** extends substantially the whole length of the underride guard **100** and provides lower support for each stabilizer **102**. Each stabilizer **102** can be intermittently spaced apart along the first beam **104** and the second beam **106**. The stabilizers **102** are spaced apart to good effect and can be welded at their upper ends **120** to mounting brackets **150**, which are in turn bolted to the structural siderails **14** at the trailer floor-support crossmembers **16**. Preferably, the top surface **121** of the vertical section **110** is connected to the mounting bracket **150**. A spacer **156** can be placed between the mounting bracket **150** and the siderail **14** of the trailer **10**.

Similarly, the inboard end **132** of the horizontal section **112** and/or the upper-inboard end **142** of the diagonal section **114** can be operatively connected to the second beam **106**. When the underride guard **100** is properly installed, the second beam **106** is positioned on the inboard side **103** of the underride guard **100** in a horizontal orientation generally parallel to the longitudinal centerline L of the floor **12**. The second beam **106** extends substantially the whole length of the underride guard **100** and provides upper support for each stabilizer **102**. Each stabilizer **102** can be intermittently spaced apart along the second beam **106**. As discussed further below, the second beam **106** also buttresses against the floor-support crossmembers **16** of the trailer **10** when the underride guard **100** is impacted laterally from the outboard side **101**.

Mounting brackets **150** can be positioned at the upper end **120** of the vertical section **110** to fasten the stabilizer **102** to the trailer **10**. In the preferred embodiment, the mounting bracket **150** is an angled mounting bracket. The preferred mounting bracket **150** is formed by elongated flanges **152**, **154** forming approximately a 90 degree angle. In other words, the first elongated flange **152** and second elongated flange **154** can have flat surfaces that are perpendicular to each other. The mounting bracket **150** can then be connected to a corner of the floor **12** at the structural siderails **14**. For example the first elongated flange **152** can connect to the side of the structural siderail **14** and the second elongated flange **154** can connect to the bottom of the floor **12** or the floor-support crossmember **16**. As such, the second elongated flange **152** can be connected to the top surface **121** of the vertical section **110**. Preferably, each stabilizer **102** is operatively connected to one mounting bracket **150**.

In this configuration, the underride guard **100** is attached to the trailer **10** along a single line of attachment along the lateral side of the trailer **10**. Furthermore, there is no physical connection of the side underride guard **100** to inboard portions of the trailer **10**. This configuration allows the side underride guard **100** to swing medially and upwardly towards the longitudinal centerline L of the trailer **10**.

For further stabilization, a cross brace or longitudinal brace **160** is operatively connected to the first beam **104** in between two stabilizers **102***a*, **102***b* to resist in-plane shear forces on the underride guard **100**. As such, the longitudinal brace **160** is configured to extend in the direction of the longitudinal centerline L of the trailer **10** when the side underride guard **100** is properly installed on the trailer **10**. Preferably, the longitudinal brace **160** comprises a first end **162** and a second end **164** opposite the first end **162**. The first end **162** of the longitudinal brace **160** can be operatively connected to the upper end **120** of a first vertical section **110***a* of a first stabilizer **102***a* or to the floor **12**, siderail **14**, or floor-support crossmember **16**, and the second end **164** can be operatively connected to the first beam **104** adjacent to the lower end **122** of a second vertical section **110***b* of a second stabilizer **102***b*. In some embodiments, one longitudinal brace **160** can be in between each pair of stabilizers **102**. In some embodiments, only one longitudinal brace **160** may be required, for example, in between the last two rear stabilizers **102***a*, **102***b*. In some embodiments, two longitudinal braces **160** can be used in between a pair of stabilizers forming an X-configuration when viewed from the side. The longitudinal brace **160** resists any longitudinal forces that could cause the underride guard **100** to "matchbox" and collapse in the longitudinal direction. In addition, the beams **104**, **106** offer torsional stiffness, which can distribute impact loads to adjacent longitudinal braces **160** of the underride guard **100**.

With reference to FIG. **6**, in the preferred embodiment a pair of underride guards **100***a*, **100***b* are operatively connected to a trailer **10** on opposite lateral sides of the trailer **10**. As such, the second underride guard **100***b* can be essentially a mirror image of the first underride guard **100***a* with the same features described above. The underride guards **100***a*, **100***b* are attached to the trailer **10** via the mounting brackets **150** so that the underride guards **100***a*, **100***b* hang below the floor **12**. In the preferred embodiment, the first underride guard **100***a* and the second underride guard **100***b* can function independently of the other. In other words, the first underride guard **100***a* and the second underride guard **100***b* are not directly connected to each other by any kind of crossmember. In some embodiments, the two underride guards **100***a*, **100***b* can be connected to each other.

In the preferred embodiment, the vertical sections **110***a*, **110***b* of each underride guard **100***a*, **100***b* have an angle of inward displacement A in which the vertical sections **110***a*, **110***b* are angled medially towards the longitudinal centerline L of the floor **12**. As such, the vertical sections **110***a*, **110***b* of the stabilizers on the first underride guard **100***a* are non-parallel to the vertical sections **110***b* of the corresponding stabilizer on the second underride guard **100***b* on the opposite lateral side of the trailer **10** because they bend toward each other. As such, the side underride guards **100***a*, **100***b* are canted inwardly towards each other.

In other words, referring to one side (here, left side of FIGS. **6** and **7**), an imaginary vertical line B extends perpendicularly from the side of the floor **12** to the ground from the lateral side of the floor **12** adjacent to the outboard side **101** of the side underride guard **100***a*. In trailers **10** with a side skirt **26**, the imaginary line B can be within the plane of the side skirt **26**. The imaginary line B can extend parallel to the flat surface of the first elongated flange **152** on the outboard side of the vertical section **110***a* and form an angle of inward displacement A of about 5 degrees to about 35 degrees with the vertical section **110***a*. In some embodiments, the angle of inward displacement A can be about 10 degrees to about 30 degrees. In some embodiments, the angle of inward displacement A can be about 15 degrees to about 25 degrees. Therefore, the vertical section **110** is non-parallel to the flat surface of the first elongated flange **152**, the siderail **14** of the trailer **10** to which the first elongated flange **152** is attached, or the side skirt **26**. This configuration can be adopted on the opposite side with the second underride guard **100***b* but with the angle of displacement in the opposite direction so that the vertical sections **110***a*, **110***b* angle towards each other.

Having the underride guards **100***a*, **100***b* angled inwardly (or medially toward the longitudinal centerline L) enables vertical trailer aerodynamic side skirts **26**, attached near the siderail **26**, to flex inboard when making contact with the ground and thus reduce damage. In addition, with the side underride guards **100***a*, **100***b* canted inwardly, the attached aerodynamic side skirt **26** needs to flex less to achieve the same horizontal orientation, and is less likely to flex outward, thereby lessening the damage to the side skirt **26**. The inward cant of the underride guards **100***a*, **100***b* and the greater ground clearance (compared to prior art designs) minimize high-centering conditions that plague other models and minimize damage to aerodynamic devices that are attached to the guard by allowing it to flex inwardly more easily.

The inward cant and movement of the underride guards **100***a*, **100***b* can be achieved because the underride guards **100***a*, **100***b* do not attach to the floor-support crossmembers **16** of the trailer **10** on the inboard side. Rather, the underride guards **100***a*, **100***b* are only attached to the floor **12** at the lateral sides as shown in FIG. **8**. The underride guards **100***a*, **100***b* hang sufficiently below the floor **12** and floor-support crossmembers **16** of the trailer **10** such that even with the inward angle of the vertical sections **110***a*, **110***b*, the second horizontal beams **106***a*, **106***b* maintain a gap **170***a*, **170***b* from the bottom flange **22** of the floor-support crossmembers **16**, even when the trailer **10** experiences deflection or bowing due to floor loads. Prior art guards that are attached to the floor-support crossmembers **16** can break when the trailer **10** is loaded due to the rigidity of the connection to the floor-support crossmember **16**, which can deflect or bow with the application of the load. Because the underride guards **100***a*, **100***b* are not connected to and are spaced apart from the floor-support crossmembers **16** of the trailer **10**, the floor **12** and floor-support crossmembers can deflect and bow independent of the underride guards **100***a*, **100***b*, and maintain the gap **170***a*, **170***b*.

In general, the gap **170***a*, **170***b* between the second horizontal beam **106** and the bottom flange **22** of the floor-support crossmember **16** can be about 0.5 inch to about 5 inches when the trailer **10** is unloaded. Preferably, the gap **170***a*, **170***b* can be about 0.75 inch to about 4 inches when the trailer **10** is unloaded. More preferably, the gap **170***a*, **170***b* can be about 1 inch to about 3 inches when the trailer is unloaded. For example, the gap **170***a*, **170***b* can be about 1.25 inch to about 2 inches when the trailer **10** is unloaded. In another example, the gap **170***a*, **170***b* can be about 0.75 inch to about 1.25 inch when the trailer **10** is unloaded. These gap **170***a*, **170***b* sizes can keep clearance between the second horizontal beam **106** and the bottom flange **22** of the floor-support crossmember **16** whether the trailer **10** is loaded or unloaded.

Because the side underride guard **100** hangs from the upper ends **120** of the vertical sections **110** of each stabilizer **102** only at the outboard side **101**, the side underride guard **100** is able to move medially toward the longitudinal centerline L of the trailer **10** (or in the inboard direction). When a lateral (impact) load is applied toward the trailer **10** on the side in which the side underride guard **100***a* is mounted (for example, the left side in FIG. **6**), the side underride guard **100** will initially rotate about its mounting bracket **150***a*. This lateral load is the first reaction load originating from the mounting bracket **150** connection to the siderail **14** of the trailer **10**. As shown in FIGS. **6**-**7**, the first beam **106***a* is deflected inward causing the second beam to deflect upward and contact the bottom flange **22** of at least one of the floor-support crossmembers **16**. As such, this rotation mechanism allows for the closing of the gap **170***a* between one or more floor-support crossmembers **16** of the trailer floor **12** and the second horizontal beam **106***a*, but only on the side of the impact. As the force of the impact pushes the second horizontal beam **106***a* into the bottom flanges **22** of the floor-support crossmember **16** of the trailer **10**, the vertical reaction load from the floor-support crossmembers **16** to the second horizontal beam **106***a* causes the bottom flanges **22** of the floor-support crossmembers **16** to locally deflect upward causing the underride guard **100** to deform the floor-support crossmember **16**, thereby forming a saddle **24** (see, FIG. **7**). The bottom flange **22** initially deflects elastically and subsequently deforms plastically around the second beam in a concave downward manner to form the saddle **24** around the second beam, thereby preventing lateral sliding of the second beam along the bottom flange beyond frictional resistance, resulting in additional lateral support to the side underride guard **100***a* that was not present prior to the contact of the second beam with the bottom flange **22**. Therefore, this contact of the second beam **106** against the bottom flange **22** of the floor-support crossmember **16** provides the second reaction load to the force that pushed the first beam inboard (first reaction load). This saddle **24**, and the strength of the floor-support crossmember **16**, prevent relative movement and restrict further rotation of the underride guard **100***a*, limiting additional travel under the trailer **10** resulting in one or more vertical down load reactions applied to the side underride guard **100***a*. This deflection creates a depression that enables a new lateral reaction load opposite to the applied lateral load.

In other words, as the second reaction load increases in magnitude in mostly a vertical direction, the bottom flanges **22** of the floor-support crossmembers **16** will deform upward (approximately) and form a saddle **24** around the contact area of the second beam **106** as shown in FIG. **7**. The deformation of the bottom flanges **22** will progress until it provides a horizontal outboard reaction load in addition to the initial vertical reaction to supplement the bracket attachment and enables the impact vehicle to lift the loaded trailer. In some embodiments, the second reaction load may compromise the web **18** of the floor-support crossmembers **16**. To minimize damage or yielding of the web **18** to the second reaction load, a reinforcing plate **180** can be installed on the web **18** of the floor-support crossmember **16** as shown in FIGS. **9**-**10**.

With reference to FIGS. **11**A-**11**C, the reinforcing plate **180** can be a flat plate. In the preferred embodiment, the reinforcing plate **180** is generally rectangular in shape. However, other shapes can be used as well, such as circular, oval, triangular, and any other polygonal shape. In general, the reinforcing plate **180** can be described as having a first surface **181** and a second surface **183** opposite the first surface **181**, the first and second surfaces **181**, **183** being bound by a top side **182**, a bottom side **184** opposite the top side **182**, and two opposing lateral sides **186**, **188** adjacent to the top side **182** and the bottom side **184**. The defining feature of the reinforcing plate **180** is a notch **190** created on the bottom side **184**. In the preferred embodiment, the notch **190** is in the form of an inverted "V" shape, although other shapes can be used, such as an inverted "U", "Z" shape, "S" shape, and the like. Essentially, the notch **190** creates a relief along the bottom side **184** of the reinforcing plate **180**.

The height H of the reinforcing plate **180** (distance from the top side **182** to the bottom side **184**) must be sufficient to fit within the top flange **20** and bottom flange **22** of the floor-support crossmember **16**. Preferably, the reinforcing plate **180** generally occupies approximately 90 percent to approximately 100 percent of the height of the web **18** (distance between the top flange **20** and the bottom flange **22**) except for the notch **190**. In some embodiments, the reinforcing plate **180** can occupy approximately 80 percent or more of the height of the web **18**. In some embodiments, the reinforcing plate **180** can occupy approximately 70 percent or more of the height of the web **18**. In some embodiments, the reinforcing plate **180** can occupy approximately 60 percent or more of the height of the web **18**. In some embodiments, the reinforcing plate **180** can occupy approximately 50 percent or more of the height of the web **18**.

The thickness T (distance from first surface **181** to the second surface **183**) of the reinforcing plate **180** can range from about 0.1 inch to about 1 inch. Preferably, the thickness T can range from about 0.20 inch to about 0.8 inch. More preferably, the thickness T can be about 0.25 inch to about 0.75 inch. For example, the thickness T can be about 0.4 inch to about 0.6 inch. Depending on the thickness T of the reinforcing plate **180** a single reinforcing plate **180** can be used on one side of the web **18**. In some embodiments, two reinforcing plates **180** can be used on opposite sides of the web **18**. If two reinforcing plates **180** are used on opposite sides of the web **18**, the notches **190** of both reinforcing plates **180** should be aligned.

The width W (distance between lateral sides **186**, **188**) of the reinforcing plate **180** can vary, but is preferably less than 10 inches. More preferably the width W of the reinforcing plate is less than about 8 inches. For example, the width W can be about 3 inches to about 5 inches.

The area of relief created by the notch **190** can be about 1 percent to about 40 percent of the surface area of the reinforcing plate **190**. Preferably, the area of relief created by the notch **190** is about 5 percent to about 30 percent of the surface area of the reinforcing plate **190**. More preferably, the area of relief created by the notch **190** is about 10 percent to about 20 percent of the surface area of the reinforcing plate **190**.

The reinforcing plate **180** can be installed on the web **18** of the floor-support crossmember **16** such that when a lateral load is applied sufficiently on the outboard side **101** of the underride guard **100** causing the second horizontal beam **106** to contact the bottom flange **22** of the floor-support crossmember **16**, the notch **190** is aligned with or corresponds with the contact point of the horizontal beam **106** and the bottom flange **22** as shown in FIG. **10**. In other words, the reinforcing plate **180** is positioned such that the notch **190** aligns with or corresponds with the anticipated location of the saddle **24**. This configuration of the reinforcing plate **180** on the web **18** of the floor-support crossbeam **16** allows the bottom flange **22** to deform while maintaining the integrity of the web **18**.

In general, the underride guard **100** configuration ensures that the braces, i.e., the vertical section **110**, horizontal section **112**, and diagonal section **114** are sufficiently short in length, and form a triangle shape, to prevent buckling under load. For example, at its longest point, the vertical section **110** can have a length of about 20 inches or less given its cross-sectional geometric properties. Preferably, the length of the vertical section **110** can be about 17 inches or less. More preferably, the length of the vertical section can be about 15 inches or less. In one example, the vertical section **110** has a length of about 15.375 inches. The horizontal section **112** can have a length of about 15 inches or less. Preferably, the horizontal section **112** can have a length of about 12 inches or less. More preferably, the horizontal section **112** can have a length of about 10 inches or less. In one example, the horizontal section **112** has a length of about 10.875 inches. The diagonal section **114** can have a length of about 23 inches or less. Preferably, the horizontal section **114** can have a length of about 20 inches or less. More preferably, the diagonal section **114** can have a length of about 17 inches or less. In one example, the diagonal section has a length of about 17.5 inches. If or when the side underride guard **100** detaches from the mounting bracket **150** or the trailer **10**, these short sections **110**, **112**, **114** will allow the triangular stabilizer **102** to roll while the three sections **110**, **112**, **114** remain attached to the first and second horizontal beams **104**, **106**.

Because the side underride guards **100***a*, **100***b* on the two opposing sides of the trailer **10** are two separate pieces, a substantial space S in the middle of the trailer **10** is created. The space S between two opposing underride guards **100***a*, **100***b* is designed to clear any trailer suspension, articulating brake hoses, and the like. This space S, created by having two separate underride guards **100***a*, **100***b* also reduces the overall weight of the underride guard **100** compared to prior art underride guards such as that disclosed in U.S. Pat. No. 11,691,583 by approximately 180 pounds. Having detached and lighter underride guards **100***a*, **100***b* simplifies installation by reducing the need for careful alignment of the two sides of the guards **100***a*, **100***b* to make sure they align with each other.

Detached and independent underride guards **100***a*, **100***b* also allow for installation variations on the two different sides. In other words, installation of the two underride guards **100***a*, **100***b* need not be mirror image installations. For example, it is possible to have an underride guard **100***a* on the left-hand side (driver side or road side) of the trailer **10** that is mounted at its forward location immediately in back of a refrigeration-unit fuel tank, while the underride guard **100***b* on the right-hand side (passenger side or curb side) of the trailer **10** is mounted immediately behind the landing legs. This design allows better access to the fuel tank and is not possible with the design that involves cross pieces that tie the two sides of the guards together.

Additionally, this space S in between the two side underride guards **100***a*, **100***b* allows for other optional trailer devices and working equipment, such as tire carriers, belly boxes, lift ramps, toolboxes, auxiliary batteries, and the like to be attached to the trailer **10** without any hindrance. Another benefit to the significant space S between the two opposing underride guards **100***a*, **100***b* is that less snow and ice can collect, form, and add significant weight to the trailer in cold environments.

The components described herein can be operatively connected together in a structurally sound manner, for example, by welding, with nuts and bolts, and the like. All measurements are provided as approximations to account for manufacturing tolerances.

In use, a side protection system can be configured on a trailer **10** by providing a first side underride guard **100***a*, comprising a first beam **104**, a second beam **106** parallel to the first beam **104**, and a series of repeating stabilizers **102** mounted on the first beam **104** and the second beam **106**. Each stabilizer **102** is intermittently spaced apart along the first beam **104** and the second beam **106**. Each stabilizer **102** in the series of repeating stabilizers comprises three sections forming a triangular shape, wherein the three sections comprise a first (vertical) section **110**, a second (horizontal) section **112** having an outboard end **130** and an inboard end **132**, the outboard end **130** connected to the first section **110**, and a third (diagonal) section **114** connected to the first section **110** and the second section **112**, wherein the first beam **104** is adjacent to where the first section **110** connects to the third section **114**, and the second beam **106** is adjacent to where the second section **112** connects to the third section **114**. A top side **134** of the second section **112** at the outboard end **130** is offset from a top surface **121** of the first section **110**.

The first side underride guard **100***a* can be connected to a first side of the trailer **10**, the trailer **10** having a longitudinal centerline L such that the first side underride guard **100***a* is configured to move medially toward the longitudinal centerline L of the trailer **10**. Preferably, each first section **110** is non-parallel to a siderail **14** of the trailer **10**, thereby forming an angle of inward displacement A towards the longitudinal centerline L. Preferably, the first side underride guard **100***a* can be connected to the trailer **10** with a plurality of mounting brackets **150**, wherein one mounting bracket **150** is operatively connected to each stabilizer **102**. Specifically, a first flange **152** can be connected to the siderail **14** of the trailer, and the second flange **154** can be connected to the first section **110**.

In this configuration, the first beam **104** is positioned on an outboard side **101** of the trailer **10** and extends longitudinally parallel to the longitudinal centerline L of the trailer **10**, and the second beam **106** is position on an inboard side **103** relative to the first beam **104** and extends longitudinally parallel to the longitudinal centerline L of the trailer **10**. The second beam **106** is configured to abut against the floor **12** or the floor-support crossmember **16** of the trailer **10** when the first underride side guard **100***a* moves medially toward the longitudinal centerline L of the trailer **10**, for example, when impacted by an automobile.

A second side underride guard **100***b* can be connected to a second side of the trailer **10** opposite the first side, the second side underride guard **100***b* being configured to move medially toward the longitudinal centerline L of the trailer **10** independent of the first side underride guard **100***a*. As such, the second side underride guard **100***b* can be a mirror image of the first underride guard **100***a*; however, because the two are not connected to each other, impact on one does not necessarily have an impact on the other.

No standard for side guard bars has been established by NHTSA regulation, understanding that the NHTSA regulations for rear underride guards are intended for the underride bar to meet an Insurance Institute for Highway Safety (IIHS) standard that is low enough to engage the structural front end of a vehicle. Side underride guards that meet the current ground clearance standards for rear underride guards repeatedly face the problem of high centering of the trailer **10** on such a side guard bar, causing damage to the side guard bar, particularly when the trailer **10** traverses changes in grade, such as those encountered on steep driveways, loading docks, railroad tracks, and the like. However, confirmed by testing, the side underride guard **100** of the present invention, installed on the side of a trailer **10** at a nominal effective ground clearance of about twenty-seven inches has been found to arrest an IIHS standard test automobile by engaging first the sheet metal and then the structure of the automobile below the windshield even though the structural front end of the vehicle is not engaged. Thus, the present invention is believed to be the first side underride guard **100** to demonstrably protect against passenger-compartment intrusion in a side impact while avoiding damage to the trailer **10** and side underride guard **100** associated with high centering impacts that are expected in normal operation of the trailer **10**.

To achieve this effective ground clearance, the lowest point of the first horizontal beams **104** can be located about 16 to 19 inches below the structural siderails **14** of a typical trailer **10** at least at their highest point where high centering is most problematic. More preferably, the first horizontal beams **104** are to be located about 17 to 18 inches below the structural siderails **14**. Effective ground clearance is to be understood in this context as measured between the bottom of the first horizontal beam **104** and a level ground surface with the trailer loaded and the suspension in the most rearward position. Variations in tire pressure, tread depth, trailer weight, load, and the like affecting trailer ground clearance are understood to be within the effective range of the nominal twenty-seven inches of effective ground clearance. In some embodiments, the first horizontal beam **104** does not need to extend fully to the wheels and tires of a trailer or tractor-trailer rig as those components are virtually impenetrable to vehicle underride. Further, the wheel assembly often is designed to slide fore and aft to accommodate various configurations, weight distributions, and highway regulations.

The side underride guards **100** are principally designed to mitigate compartment intrusion of an automobile that impacts the trailer in the space between the wheels and the drop legs. For example, the configurations described herein are designed to allow the hood of the automobile to go under the underride guard **100**, but stop before the A-pillar of the automobile, thereby preventing the passenger compartment from going under the underride guard **100**. The ground clearance of the first horizontal beams **104** may also be reduced toward the wheels and tires of some vehicles as the likelihood of high centering may diminish. However, the lower the first horizontal beam **104**, the more interference with trailer side skirt **26** flexure will be encountered.

The foregoing description of the preferred embodiment of the invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. It is intended that the scope of the invention not be limited by this detailed description, but by the claims and the equivalents to the claims appended hereto.

What is claimed is:

**1**. A side impact guard for a trailer, the side impact guard comprising a first side underride guard for mounting to a first side of the trailer, the first underride guard comprising:

a) a pair of beams comprising a first beam and a second beam parallel to the first beam;

b) a series of repeating stabilizers mounted on the pair of beams, wherein each stabilizer is intermittently spaced apart along the first beam and the second beam;

c) a longitudinal brace operatively connected to the first beam in between two stabilizers to resist in-plane shear forces on the first side underride guard longitudinal to a length of the trailer;

d) a plurality of mounting brackets operatively connected to each stabilizer, each mounting bracket configured to connect to a structural side rail of the trailer, wherein each mounting bracket comprises a first elongated flange having a flat surface and a second elongated flange having a flat surface that is perpendicular to the flat surface of the first elongated flange;

e) wherein each stabilizer in the series of repeating stabilizers is triangular in shape, each stabilizer comprising three sections, wherein when installed on the trailer, a vertical section is approximately in a vertical orientation, a horizontal section is approximately in a horizontal orientation, and a diagonal section is in a diagonal orientation, wherein the vertical, horizontal, and diagonal sections form a triangle;

wherein the vertical section comprises an upper end terminating at a top surface, and a lower end terminating at a bottom surface opposite the upper end, the horizontal section comprises an outboard end and an inboard end opposite the outboard end, and the diagonal section comprises a lower-outboard end and an upper-inboard end opposite the lower-outboard end;

f) wherein the outboard end of the horizontal section is operatively connected to the upper end of the vertical section, the lower-outboard end of the diagonal section is operatively connected to the lower end of the vertical section, and the inboard end of the horizontal section is operatively connected to the upper-inboard end of the diagonal section,

g) wherein a top side of the horizontal section at the outboard end is offset from the top surface of the vertical section by about 1 inch to about 5 inches,

h) wherein the lower end of the vertical section is operatively connected to the first beam, and the lower-outboard end of the diagonal section is operatively connected to the first beam, thereby positioning the first beam on an outboard side of the first side underride guard;

i) wherein the inboard end of the horizontal section is operatively connected to the second beam, and the upper-inboard end of the diagonal section is operatively connected to the second beam, thereby positioning the second beam on an inboard side of the first side underride guard;

j) wherein the longitudinal brace comprises a first end and a second end opposite the first end, wherein the first end of the cross brace is operatively connected to the upper end of the vertical section of a first stabilizer, and the second end is operatively connected to the first beam adjacent at the lower end of the vertical section of a second stabilizer; and wherein each vertical section is connected to a respective mounting bracket at the flat surface of the second elongated flange, non-parallel to the flat surface of the first elongated flange of the respective mounting bracket.

**2**. The side impact guard of claim **1**, further comprising a second side underride guard configured as a mirror image of the first underride guard for mounting on a second side of the trailer opposite the first side of the trailer with a space therebetween, wherein the first underride guard is configured to function independently of the second underride guard.

**3**. The side impact guard of claim **2**, wherein the vertical sections of each underride guard have an angle of inward displacement in which the vertical sections are angled medially towards the center of the trailer.

**4**. An underride guard for mounting to a side of a trailer, the underride guard comprising:

a) a first beam;

b) a second beam parallel to the first beam;

c) a series of repeating stabilizers mounted on the first beam and the second beam, wherein each stabilizer is intermittently spaced apart along the first beam and the second beam; and

d) a plurality of mounting brackets operatively connected to each stabilizer;

e) wherein each stabilizer in the series of repeating stabilizers comprises three sections forming a triangular shape, wherein the three sections comprise a vertical section having an upper end and a lower end, a horizontal section having an outboard end and an inboard end, the outboard end connected to the upper end of the vertical section, and a diagonal section connected to the lower end of the vertical section, and the diagonal section connected to the inboard end of the horizontal section, wherein the first beam is adjacent to where the vertical section connects to the diagonal section, and the second beam is adjacent to where the horizontal section connects to the diagonal section;

f) wherein a top side of the diagonal section at the outboard end is offset from a top surface of the vertical section; and

g) wherein each mounting bracket is configured for connecting one of the stabilizers to the side of the trailer, wherein each mounting bracket comprises a first elongated flange having a flat surface, and a second elongated flange having a flat surface that is perpendicular to the flat surface of the first elongated flange, wherein the flat surface of the second elongated flange is connected to the top surface of the vertical section of the respective mounting bracket, and wherein each vertical section is non-parallel to the flat surface of the first elongated flange of the respective mounting bracket, thereby forming an angle of inward displacement.

**5**. The underride guard of claim **4**, wherein the offset is about 1 inch to 5 inches.

**6**. The underride guard of claim **5**, wherein the angle of inward displacement is about 5 degrees to about 15 degrees.

**7**. The underride guard of claim **6**, wherein each vertical section has a length of about 15 inches to about 20 inches, each horizontal section has a length of about 10 inches to about 15 inches, and each diagonal section has a length of about 17 inches to about 23 inches.

**8**. The underride guard of claim **7**, further comprising a longitudinal brace operatively connected to the first beam in between two stabilizers.

**9**. The underride guard of claim **4**, wherein the angle of inward displacement is about 5 degrees to about 15 degrees.

**10**. The underride guard of claim **4**, wherein each vertical section has a length of about 15 inches to about 20 inches.

**11**. The underride guard of claim **4**, wherein each horizontal section has a length of about 10 inches to about 15 inches.

**12**. The underride guard of claim **4**, wherein each diagonal section has a length of about 17 inches to about 23 inches.

**13**. The underride guard of claim **4**, further comprising a longitudinal brace operatively connected to the first beam in between two stabilizers.

**14**. A method of configuring a trailer with a side impact guard, comprising the steps of:

a) providing a first side underride guard, comprising a first beam, a second beam parallel to the first beam, and a series of repeating stabilizers mounted on the first beam and the second beam, wherein each stabilizer is intermittently spaced apart along the first beam and the second beam, wherein each stabilizer in the series of repeating stabilizers comprises three sections forming a triangular shape, wherein the three sections comprise a vertical section, a horizontal section having an outboard end and an inboard end, the outboard end connected to the vertical section, and a diagonal section connected to the vertical section and the horizontal section, wherein the first beam is adjacent to where the vertical section connects to the diagonal section, and the second beam is adjacent to where the horizontal section connects to the diagonal section, wherein a top side of the horizontal section at the outboard end is offset from a top surface of the vertical section; and

b) connecting the first side underride guard to a first side of the trailer, the trailer having a longitudinal centerline such that the first side underride guard is configured to move medially toward the longitudinal centerline of the trailer.

**15**. The method of claim **14**, wherein each vertical section is non-parallel to a siderail of the trailer, thereby forming an angle of inward displacement towards the longitudinal centerline.

**16**. The method of claim **15**, wherein the first side underride guard is connected to the siderail of the trailer with a plurality of mounting brackets, wherein one mounting bracket is operatively connected to each stabilizer, wherein each mounting bracket comprises a first elongated flange having a flat surface, and a second elongated flange having a flat surface that is perpendicular to the flat surface of the first elongated flange, wherein the flat surface of the second elongated flange is connected to the top side of the vertical section of the respective mounting bracket.

**17**. The method of claim **16**, wherein the first beam is positioned on an outboard side of the trailer and extends longitudinally parallel to the longitudinal centerline of the trailer, wherein the second beam is position inboard relative to the first beam and extends longitudinally parallel to the longitudinal centerline of the trailer, wherein the second beam is configured to abut against a floor-support cross-members of the trailer when the first underride side guard moves medially toward the longitudinal centerline of the trailer.

**18**. The method of claim **17**, wherein the first horizontal beam is positioned about sixteen to nineteen inches below a structural siderail of the trailer where the mounting brackets are mounted.

**19**. The method of claim **18**, further comprising connecting a second side underride guard to a second side of the trailer opposite the first side, the second side underride guard configured to move medially toward the longitudinal centerline of the trailer independent of the first side underride guard.

**20**. The method of claim **19**, further comprising connecting a longitudinal brace to the first beam in between two stabilizers.

**21**. The method of claim **14**, wherein the second beam is gapped away from floor-support crossmembers of the trailer vertically by approximately 0.75 inch to approximately 1.5 inch to allow the floor-support crossmembers to deflect from trailer loading independently from the side underride guard.

**22**. The method of claim **21**, wherein the second beam is configured to contact the floor-support crossmembers only when the first beam is deflected inboard.

**23**. The method of claim **22**, wherein upon lateral impact of the side underride guard, the first beam is configured to deflect inboard causing the second beam to deflect upward and contact a bottom flange of at least one of the floor-support crossmembers of the trailer, said bottom flange initially deflecting elastically and subsequently deforming plastically around the second beam in a concave downward manner to form a saddle around the second beam, thereby preventing lateral sliding of the second beam along the bottom flange, beyond frictional resistance, resulting in additional lateral support to the side underride guard that was not present prior to the contact of the second beam with the bottom flange.

**24**. The method of claim **23**, further comprising the step of installing a reinforcing plate on the floor-support cross-member to supplement deforming the cross-member bottom flange to the second beam upon impact of the side underride guard.

**25**. The method of claim **24**, wherein the reinforcing plate is a flat plate comprising a top side, a bottom side opposite the top side, a first lateral side adjacent to the top side and the bottom side, a second lateral side opposite the first lateral side and adjacent to the top side and the bottom side, and a notch defined along the bottom side, wherein the reinforcing plate is installed on the floor-support crossmember such that the notch corresponds with where the second beam contacts the bottom flange upon lateral impact of the side underride guard.

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